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CSA - 1306

CO3 - 3

TASK 1 - Context free grammar design & derivation

A context-free grammar (CFG) is a formal grammar used to ~~describ~~ describe the syntax of a language. It specifies how strings can be generated.

$G = (V, T, P, S)$

$V \longrightarrow$ Non-terminals

$T \longrightarrow$ Terminals

$P \longrightarrow$ Production rules

$S \longrightarrow$ Start symbol

$V = \{E, T, F\}$

$T = \{id, +, *, (,)\}$

$P = E \longrightarrow E+T \mid T$

$T \longrightarrow T*F \mid F$

$F \longrightarrow (E) \mid id$

$S = E$

LMD :-

E
↓
E + T
↓
T + T
↓
F + T
↓
id + T
↓
a + T
↓
a + T * F
↓
a + F * F
↓
a + id * F
↓
a + b * F
↓
a + b * id
↓

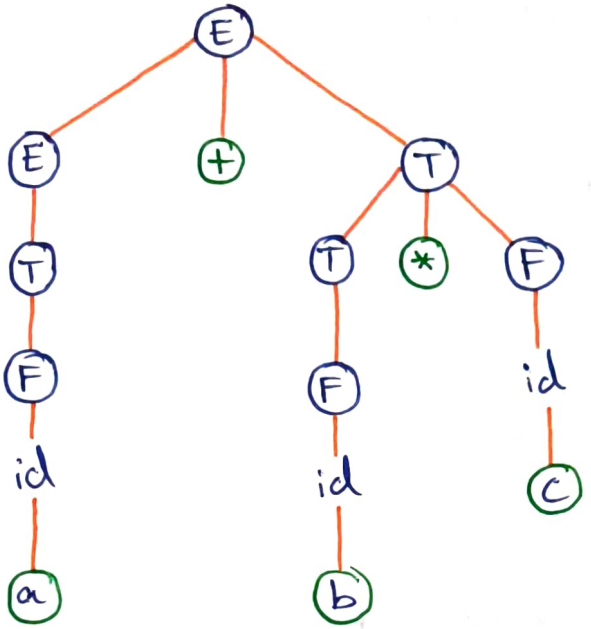
a + b * c

RMD :-

E
↓
E + T
↓
E + T * F
↓
E + T * id
↓
E + F * id
↓
E + id * id
↓
E + b * c
↓
T + b * c
↓
F + b * c
↓
id + b * c
↓

a + b * c

Parse tree :-



TASK - 2 : Simplification of context-free grammar

Given CFG :-

$S \longrightarrow AB|a$

$A \longrightarrow aA|\varepsilon$

$B \longrightarrow bB|C$

$C \longrightarrow c$

Nullable variables :- $\{A\}$

Eliminate ε-productions :-

$S \longrightarrow AB|B|a$

$A \longrightarrow aA|a$

$B \longrightarrow bB|C$

$C \longrightarrow c$

Eliminate unit productions :-

$S \longrightarrow AB \mid a \mid bB \mid c$

$A \longrightarrow aA \mid a$

$B \longrightarrow bB \mid c$

$C \longrightarrow c$

C is unreachable $\Rightarrow$ Remove it

Final simplified CFG :-

$S \longrightarrow AB \mid a \mid bB \mid c$

$A \longrightarrow aA \mid a$

$B \longrightarrow bB \mid c$

TASK 3 - Conversion to Chomsky normal form

Given CFG :-

$S \longrightarrow aAB \mid b$

$A \longrightarrow aA \mid a$

$B \longrightarrow bB \mid b$

Productions that don't satisfy CNF :-

* $S \longrightarrow aAB$
* $A \longrightarrow aA$
* $B \longrightarrow bB$

Variables for terminal symbols :-

Let $X \rightarrow a$
$Y \rightarrow b$

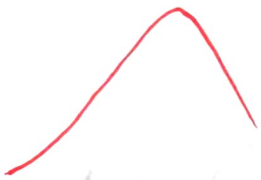
$\Rightarrow$

$S \rightarrow XAB \mid b$
$A \rightarrow XA \mid a$
$B \rightarrow YB \mid b$
$X \rightarrow a$
$y \rightarrow b$

Breaking productions :-

$S \rightarrow XAB$; Contains 3 nonterminals

$S \rightarrow XC \mid b$
$A \rightarrow XA \mid a$
$B \rightarrow YB \mid b$
$C \rightarrow AB$
$X \rightarrow a$
$y \rightarrow b$



This satisfies CNF restrictions

∴, resulting grammar is in CNF.

TASK 4 - Conversion into Greibach normal form

Given CFG :- $S \rightarrow AB \mid a$
$A \rightarrow aA \mid a$
$B \rightarrow bB \mid b$

Productions that don't satisfy GNF :- $A \rightarrow aA$

Apply required transformations.

Final GNF form :-

$S \rightarrow aAB \mid aB \mid a$
$A \rightarrow aA \mid a$
$B \rightarrow bB \mid b$

$S \rightarrow aAB$

$A \rightarrow a \Rightarrow aAB \rightarrow aaB$

$B \rightarrow b \Rightarrow aaB \rightarrow aab$

$\therefore$ $S \rightarrow aAB \rightarrow aaB \rightarrow aab$

TASK 5 - Context-free language properties and Pumping lemma

Given Language :- $L = \{a^n b^n c^n \mid n \geq 1\}$

$L = \{abc, aabbcc, aaabbbccc, \ldots\}$

Let $c = 6$

$|aaabbbccc| = 6 \geq c$

$w = \underbrace{aaa}_{x}\underbrace{bbb}_{y}\underbrace{ccc}_{z} \Rightarrow$ $x = aaa$, $y = bbb$, $z = ccc$

$|y| = |bbb| = 3 > 0$

$|xy| = |aaabbb| = 6 \leq 6$



∴ $i = 0 \Rightarrow aaaccc$ ✗ Rejected

$i = 1 \Rightarrow aaabbbccc$ ✓ Accepted

$i = 2 \Rightarrow aaabbbbbbccc$ ✗ Rejected

∴ Language is not regular.